Meeting Plans August 31 362

- SSTV figured out
- IQ sampling
- Things needed to research for GPS
    - Demodulation: Aaryan
        - Libraries
        - Algorithms
    - Decoding: Michael
        - Libraries
        - Algorithms
            - CDMA
            - Hamming Code
            - Interleaving
    - Binary format
        - Protocols
        - Algorithms
        - Libraries
    - How RadioSConde Transmit
    - USB interface for the NOOELEC

- Ideas
    - Heatmap for the gps tracker. Multiplexed frequency receive with scan

Component Documentation and Nomenclature

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1. Receiving

400-406 MHz frequency band

Transmission Speeds

- RS41: Less than 1s (Continuous)
- DFM-17: Less than 1s (Complete packet)

~ 24 bytes/second
Usually fly for 2+ hours <- and fly about 2 times a day

2. Demodulation

I/Q Conversion

Amplitude: $|I + jQ| = \sqrt{I^2 + Q^2}$

Phase: $\phi = \arctan(Q/I)$

Frequency: Changes in phase over time

Extracting Data from its I/Q representation

Librtlsdr library - raw I/Q capture
https://github.com/librtlsdr

Software Interfaces - Demodulation

Soapy SDR

https://github.com/pothosware/SoapySDR/wiki

Gnu Radio Demod

https://github.com/gnuradio/gnuradio/blob/master/gr-digital/python/digital/gfsk.py

SondeHub Radiosonde Decoding Software (Most common RTL-SDRs RS41, iMS-100)
In python though - not sure if for embedded systems
https://github.com/projecthorus/sondehub-decoders

Raspberry Pico Demod (DSP)

PicoDuck

https://github.com/playduck/pico-dsp

Liquid DSP

https://pypi.org/project/liquid-dsp/

Documentation

- https://www.reddit.com/r/RTLSDR/comments/wyarqp/433mhz_rf_receiver_for_software_demodulation_on/
- DFM-17 Radiosonde
- DFM-17 Sigid Wiki Page

- DFM-17 Radiosonde Datasheet
    - Tuning range: 400 - 405.99 MHz
    - Max. range: > 250 km
    - Bandwidth: < 12 kHz
    - Data downlink: 1250 bit/s
- Si4063
- RS41_Decoding
- Radiosonde_auto_rx
- Radiosonde documentation
- NOOELEC Datasheet
    - 0.5PPM TCXO
    - 0.1 to 25MHz direct sampling tuning range
    - 25 to 1750MHz quadrature sampling tuning range

Receiver IC:

https://www.silabs.com/documents/public/data-sheets/Si4355.pdf

USB HOST:
https://www.analog.com/media/en/technical-documentation/data-sheets/MAX3421E.pdf